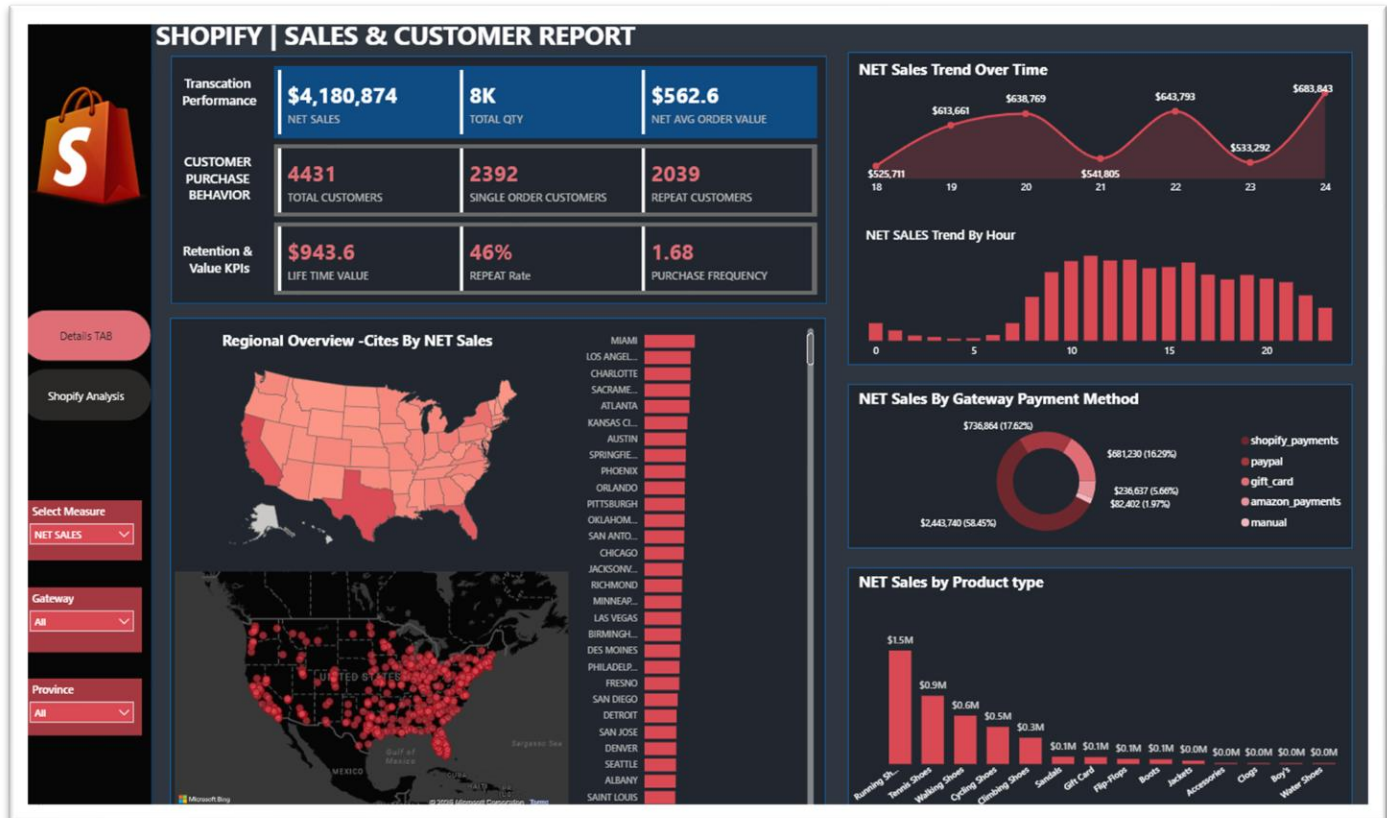


Shopify Sales Analysis- Business Intelligence Report

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Final Dashboard View



Stage 0: Problem Statement

Project Goal & Stakeholders: The objective of this project is to transform raw Shopify transactional data into an interactive intelligence tool. The primary stakeholders are e-commerce managers, marketing directors, and inventory planners who need to spot patterns in revenue generation and customer retention to make data-driven decisions.

- **Five Business Questions:**

1. What specific product categories drive the most revenue?
2. At what time of day do customers buy the most, and how can marketing leverage this?
3. What is the ratio of new versus returning customers?
4. What is the average estimated lifetime value of our customers?
5. What is our baseline repeat-purchase rate?

- **Key KPIs Tracked:** Total Revenue, Total Orders, Average Order Value (AOV), Repeat-Purchase Rate, and Customer Lifetime Value (CLV).
- **Main Objective & Steps:** The single deliverable is a fully interactive Power BI dashboard. The three steps to reach this were: 1) Extracting and cleaning data via Power Query, 2) Designing a relational Star Schema data model, and 3) Engineering DAX measures to power the visualizations.

Stage 1: Data Preparation & Modeling

1.1 Import & Clean:

- The raw Shopify dataset was imported into Power Query, consisting of one primary table with over 7,400 rows. Key columns included Customer Id, Total Price Usd, Invoice Date, and Product Type. Cleaning steps included enforcing strict currency data types for financials, capitalizing location strings for standardization, and filtering out null product identifiers.
- **Helper Columns:** To analyze purchasing behavior by time, an Hour helper column was extracted from the Invoice Date timestamp.

1.2 Data Model:

1. The architecture utilizes a Star Schema. A dedicated Dim_Date calendar table was generated and marked as a date table. This was physically linked to the sales fact table via a 1-to-Many relationship, which is critical for enabling accurate time-based DAX calculations.

Stage 2: Transaction Performance Analysis

- **DAX Measures:**
 - **Total Revenue** = *SUM('shopify_sales'[Total Price Usd])* — Calculates the gross income generated by the store.
 - **Total Orders** = *DISTINCTCOUNT('shopify_sales'[Order Number])* — Counts the unique transaction receipts to avoid double-counting multi-item carts.
 - **AOV** = *DIVIDE([Total Revenue], [Total Orders], 0)* — Calculates the average monetary spend per individual order.
- **Trends & Performance:** Revenue fluctuates daily with notable peaks around the 20th of the month. Running Shoes and Tennis Shoes are the dominant revenue drivers, while sandals and water shoes severely underperform.
- **Time of Day:** Order volume spikes drastically between 10:00 AM and 2:00 PM. The store can use this data to trigger targeted mid-day marketing campaigns.

Stage 3: Customer Purchasing Behaviour

- **DAX Measures:**
 - **Total Customers** = **DISTINCTCOUNT('shopify_sales'[Customer Id])** — Counts the unique number of individual buyers.
 - **Repeat Customers** = **COUNTROWS(FILTER(VALUES('shopify_sales'[Customer Id]), CALCULATE(DISTINCTCOUNT('shopify_sales'[Order Number])) > 1))** — Identifies buyers who have placed more than one order.
 - **Single Order Customers** = **[Total Customers] - [Repeat Customers]** — Calculates the volume of one-time buyers.
- **Behavioral Split:** Out of 4,431 unique customers, 2,039 are returning buyers, highlighting a highly engaged customer base

Order Number	Customer Name	Customer Id	Province	CITY	Zip Code	Product Type	Gateway	NET SALES	Total Tax	Total Price Us
910	Maure Patek	1	Texas	ABILENE	73655	Running Shoes	shopify_payments	\$497	49.71	546.8
927	Latia Birkwood	1	Texas	ABILENE	62906	Running Shoes	paypal	\$518	51.75	569.2
1038	Joan Casa	1	Texas	ABILENE	39466	Running Shoes	gift_card	\$487	48.67	535.4
3410	Orelee McGoogan	1	Texas	ABILENE	8452	Tennis Shoes	gift_card	\$537	53.71	590.8
3532	Estrella Giacobilo	1	Texas	ABILENE	12502	Tennis Shoes	shopify_payments	\$602	60.23	662.5
5468	Abey O'Kennedy	1	Texas	ABILENE	2987	Gift Card	paypal	\$514	51.41	565.5
5474	Antonella Rowthorn	1	Texas	ABILENE	85253	Running Shoes	gift_card	\$580	57.99	637.9
6412	Olav Enders	1	Texas	ABILENE	96733	Running Shoes	gift_card	\$537	53.71	590.8
2541	Durward Schutte	1	South Carolina	AIKEN	48455	Walking Shoes	shopify_payments	\$811	81.11	892.2
4826	Mercie Curston	1	South Carolina	AIKEN	8266	Climbing Shoes	amazon_payments	\$811	81.11	892.2
5207	Lesly Perutto	1	South Carolina	AIKEN	16940	Tennis Shoes	paypal	\$517	51.71	568.8
101	Perice Moppett	1	Ohio	AKRON	23773	Cycling Shoes	gift_card	\$510	50.95	560.4
354	Rivy Pietersma	1	Ohio	AKRON	2482	Walking Shoes	gift_card	\$494	49.37	543.1
438	Lyndsie Deaton	1	Ohio	AKRON	77968	Climbing Shoes	shopify_payments	\$537	53.71	590.8
765	Rowena Dollen	1	Ohio	AKRON	52481	Running Shoes	gift_card	\$596	59.59	655.5
1329	Vernee Iskovic	1	Ohio	AKRON	3653	Running Shoes	shopify_payments	\$660	65.96	725.5
2187	Vida Rene	1	Ohio	AKRON	48005	Running Shoes	shopify_payments	\$615	61.51	676.6
2189	Inge Rowbotham	1	Ohio	AKRON	37270	Cycling Shoes	shopify_payments	\$625	62.51	687.6
2199	Angele Cicetti	1	Ohio	AKRON	18695	Running Shoes	paypal	\$596	59.59	655.5
2417	Clayson Medmore	1	Ohio	AKRON	35117	Running Shoes	paypal	\$625	62.51	687.6
2583	Dilan Oles	1	Ohio	AKRON	33182	Running Shoes	shopify_payments	\$510	50.95	560.4
2610	Rose Melsom	1	Ohio	AKRON	5570	Climbing Shoes	shopify_payments	\$537	53.71	590.8
2635	Judith Wareing	1	Ohio	AKRON	14869	Running Shoes	shopify_payments	\$625	62.51	687.6
3371	Ardelis Olivera	1	Ohio	AKRON	77144	Running Shoes	paypal	\$595	59.51	654.6
3932	Kakalina Ferdinand	1	Ohio	AKRON	75483	Running Shoes	shopify_payments	\$548	54.79	602.7
3948	Timofei Schimon	1	Ohio	AKRON	42701	Gift Card	shopify_payments	\$514	51.41	565.5
4148	Emelen Methven	1	Ohio	AKRON	79208	Running Shoes	shopify_payments	\$596	59.59	655.5
4549	Madelle Dowman	1	Ohio	AKRON	4000	Flip-Flops	gift_card	\$507	50.71	557.8
5466	Monie Toth	1	Ohio	AKRON	53753	Cycling Shoes	shopify_payments	\$635	63.51	698.6
5615	Valery Godlip	1	Ohio	AKRON	68272	Running Shoes	gift_card	\$504	50.41	554.5
5968	Jerrold Witham	1	Ohio	AKRON	22091	Flip-Flops	paypal	\$526	52.55	578.0
6061	Phyllida Passy	1	Ohio	AKRON	47079	Tennis Shoes	paypal	\$596	59.59	655.5
6379	Albie Getch	1	Ohio	AKRON	81151	Running Shoes	paypal	\$693	69.31	762.4
6509	Jamill McKinstry	1	Ohio	AKRON	97749	Gift Card	shopify_payments	\$514	51.41	565.5
6926	Ediva Cornillot	1	Ohio	AKRON	12980	Tennis Shoes	gift_card	\$510	50.95	560.4
73	Tiebold Swanborrow	1	New York	ALBANY	21815	Running Shoes	paypal	\$527	52.71	579.8
88	Enika Rosengren	1	Georgia	ALBANY	41644	Cycling Shoes	gift_card	\$510	50.95	560.4
Total		7431			367597147			\$4,180,874	418,087.35	4,598,960.8

Stage 4: Retention & Long-Term Customer Value

- **DAX Measures:**

- **Repeat Rate** = $DIVIDE([Repeat\ Customers], [Total\ Customers], 0)$ — Calculates the percentage share of customers who order more than once.
- **Purchase Frequency** = $DIVIDE([Total\ Orders], [Total\ Customers], 0)$ — Averages the number of distinct orders placed per customer.
- **Life Time Value** = $[AOV] * [Purchase\ Frequency]$ — Estimates the total long-term revenue a single customer brings to the business

- **Insights:** The business boasts a high retention metric with a 46% repeat purchase rate and an estimated CLV of \$943.60. The company should prioritize keeping these repeat customers through exclusive loyalty programs and re-engagement campaigns.

Stage 5: Dashboard Design & Insights

Layout Choices: The dashboard utilizes an F-pattern design, organizing the narrative top-to-bottom. Headline KPIs are positioned in the top-left for immediate visibility, followed by broader temporal trends and detailed breakdowns on the right. Interactive slicers were placed in a persistent left-hand sidebar so users can explore data by Date, Gateway, and Province.

Business Recommendations:

- **Optimize Marketing Timing:** Because purchasing volume spikes mid-day, marketing teams should schedule promotional email blasts precisely at 10:30 AM to capture this audience.
- **Reallocate Inventory Capital:** Procurement budgets should be aggressively reallocated to maintain high stock levels for Running and Tennis shoes, while defunding low-performing products like clogs.
- **Launch a VIP Program:** With 46% of customers already returning, instituting a VIP loyalty tier will help maximize the \$943 Customer Lifetime Value and prevent churn.